University of Florida, Department of Mathematics

Topology First-Year Exam, August 2021, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Is it true that every continuous map $f : [0,1] \to [0,1)$ has a fixed point. Justify your answer.

2. Show that every compact Hausdorff space is regular.

3. Is it true that

(a) every continuous injective map $f : (0,1) \to \mathbb{R}$ is an embedding?

(b) every continuous injective map $f : (0,1) \to \mathbb{R}^2$ is an embedding?

4. Show that Greek letter theta Θ is not homeomorphic to the figure eight 8. Formally, $\Theta = S^1 \cup [-1,1] \times \{0\}$ and $8 = S^1_- \cup S^1_+$ are subspaces of $\mathbb{R}^2$, where S^1_- is the unit circle centered at the point $(0,-1)$ and S^1_+ is the unit circle centered at $(0,1)$.

5. This problem has two parts.

(a) Let $A \subset \mathbb{R}^2$ be a countable set. Prove that $\mathbb{R}^2 \setminus A$ is path connected.

(b) Let $B \subset \mathbb{R}^3$ be the union of a countable family of lines in $\mathbb{R}^3$. Prove that $\mathbb{R}^3 \setminus B$ is path connected.

Answer the following with complete definitions or statements or proofs.

6. State the Extreme Value Theorem.

7. State the Tietze Extension Theorem.

8. Let $\mathbb{R}_\ell$ denote the reals with the lower limit topology. Is the subspace $\mathbb{R}_\ell \cap [0,1]$

(a) compact?

(b) connected?

9. Is every compact Hausdorff space normal?

10. Give definition of a locally connected space. Is every path connected space locally connected?